Today we will discuss two implementable classifiers.

In the discussion of optimal classifiers we assumed that we know the likelihoods.

posterior probability

$$P(\omega_1|\bar{x}) = \frac{p(\bar{x}|\omega_1)P(\omega_1)}{p(\bar{x})} \quad (1)$$

likelihood of $\bar{x}$ given ω_1

prior probability of class ω_1.

$$\vdots$$

$$P(\omega_K|\bar{x}) = \frac{p(\bar{x}|\omega_K)P(\omega_K)}{p(\bar{x})}$$

These are just applications of Bayes' Law.

If we want an optimal classifier in the sense of minimizing error we always pick the class with the greatest posterior probability. For binary classification

$$\text{Decide } \begin{cases} \omega_1 & \text{if } P(\omega_1|\bar{x}) > P(\omega_2|\bar{x}) \\ \omega_2 & \text{otherwise} \end{cases} \quad (2)$$

~~what~~ substituting (1) into (2) yields

$$\text{Decide } \begin{cases} \omega_1 & \text{if } p(\bar{x}|\omega_1)P(\omega_1) > p(\bar{x}|\omega_2)P(\omega_2) \\ \omega_2 & \text{otherwise} \end{cases} \quad (3)$$

~~If we want an optimal classifier in the sense of minimizing~~ risk (i.e. minimizing expected loss) we always choose the class with action that minimizes loss. For binary classification this becomes

$$\text{Decide } \begin{cases} \omega_1 & \text{if } P(\bar{x}|\omega_1)P(\omega_1)(\lambda_{21}-\lambda_{11}) > P(\bar{x}|\omega_2)P(\omega_2)(\lambda_{12}-\lambda_{22}) \\ \omega_2 & \text{otherwise} \end{cases} \quad (4)$$

The decision rule in (3) is for an optimal classifier because we know the likelihoods $p(\bar{x}|\omega_1)$ and $p(\bar{x}|\omega_2)$, and we know the prior probabilities $P(\omega_1)$ and $P(\omega_2)$. In practice, we do not know the likelihoods and priors.

1) we measure prior probabilities as $\frac{\#\text{ samples in training set belonging to class } y}{\#\text{ samples in training set.}}$

2) we model the likelihoods

Generative models

When we build a machine learning model that models the likelihoods, i.e., $p(\bar{x}|y)$, we call this a <u>generative model</u>. Why? If we have a good model of $p(\bar{x}|y)$ we can "generate" new samples that ~~adhere to~~ obey the same distribution.

This use of the term "generative" is a little different from the more modern usage of "generative" as applied to AI-generated art or music or to the output of chatbots based on large language models.

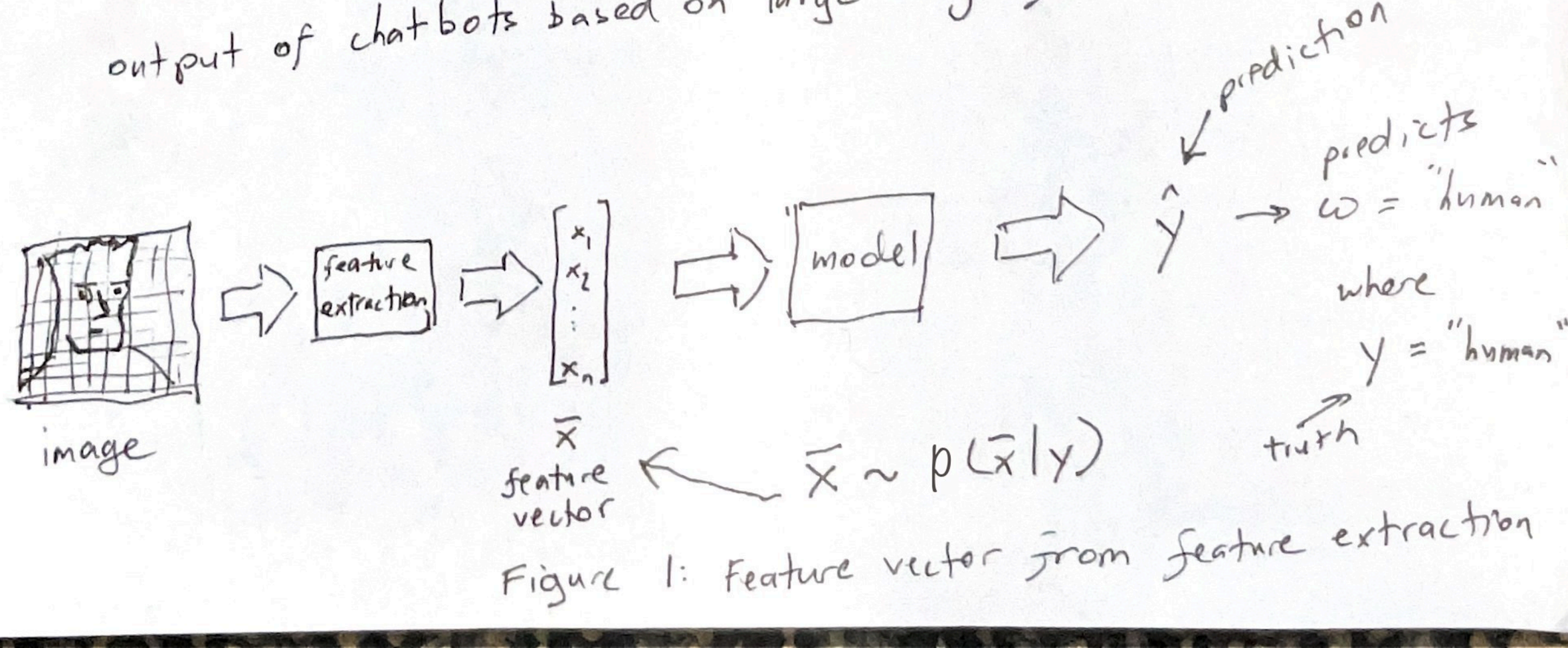


Figure 1: Feature vector from feature extraction

In the traditional sense, a generative model allows us to generate synthetic data, e.g., for <u>data augmentation</u> wherein synthetic values for $\bar{x}$ obey $p(\bar{x}|y)$ but $\bar{x}$ is a feature vector. A feature vector was extracted from ~~some other~~ a data source like an image, audio, or video.

Feature extraction often reduces something complex down to a ~~fixed~~ feature vector comprised a much smaller number of dimensions.

image $h \times w$ pixels
Each pixel has its own value and thus could be considered a dimension.

Ex: $h = 2160$, $w = 3840$
$\approx$ 8.3M dimensions
n may be 30 dimensions.

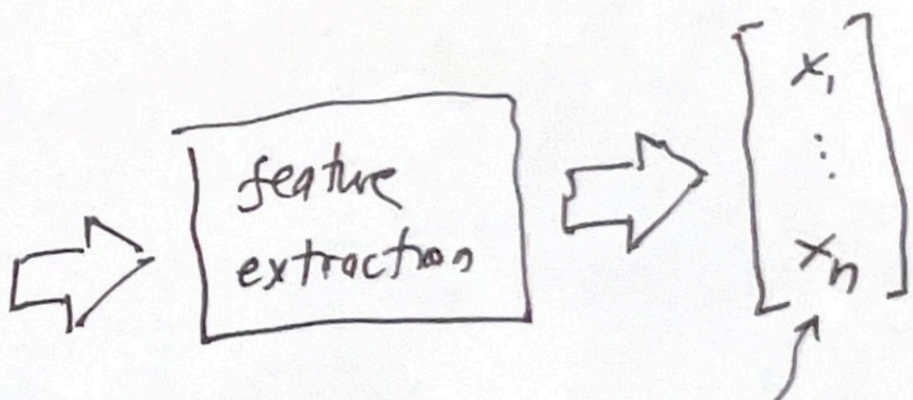


Figure 2: dimensionality reduction

~~One~~ a Feature extraction is irreversible meaning being able to generate synthetic values for $\bar{x} \sim p(\bar{x}|y)$ does not provide anything that looks like an ~~sample drawn from~~ image.

In the modern modern sense, generate AI

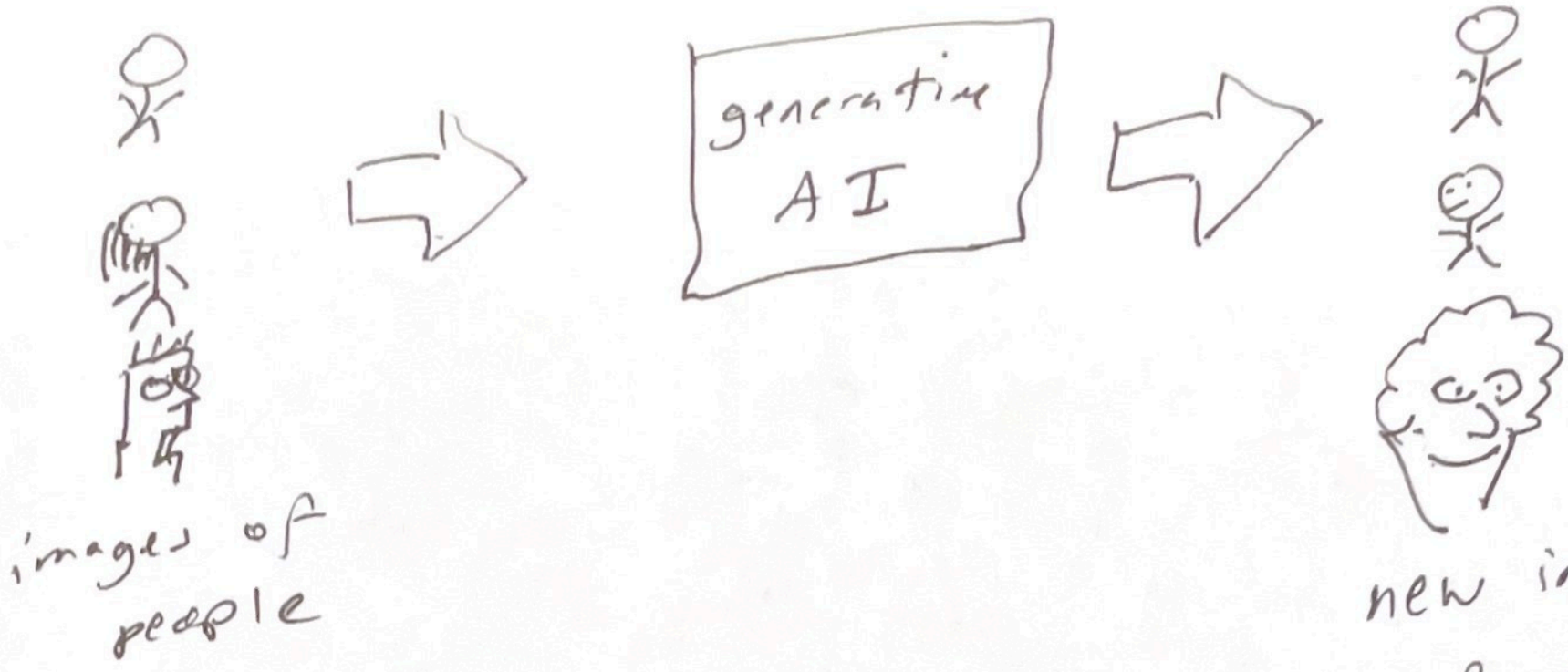


Figure 3: modern usage of the term "generative"

In this lecture we are referring to "generative" in the classical sense. We model $\bar{x} \sim p(\bar{x}|y)$

~~Generative~~

~~LDA - Gradient Gaussian Discriminant Analysis~~

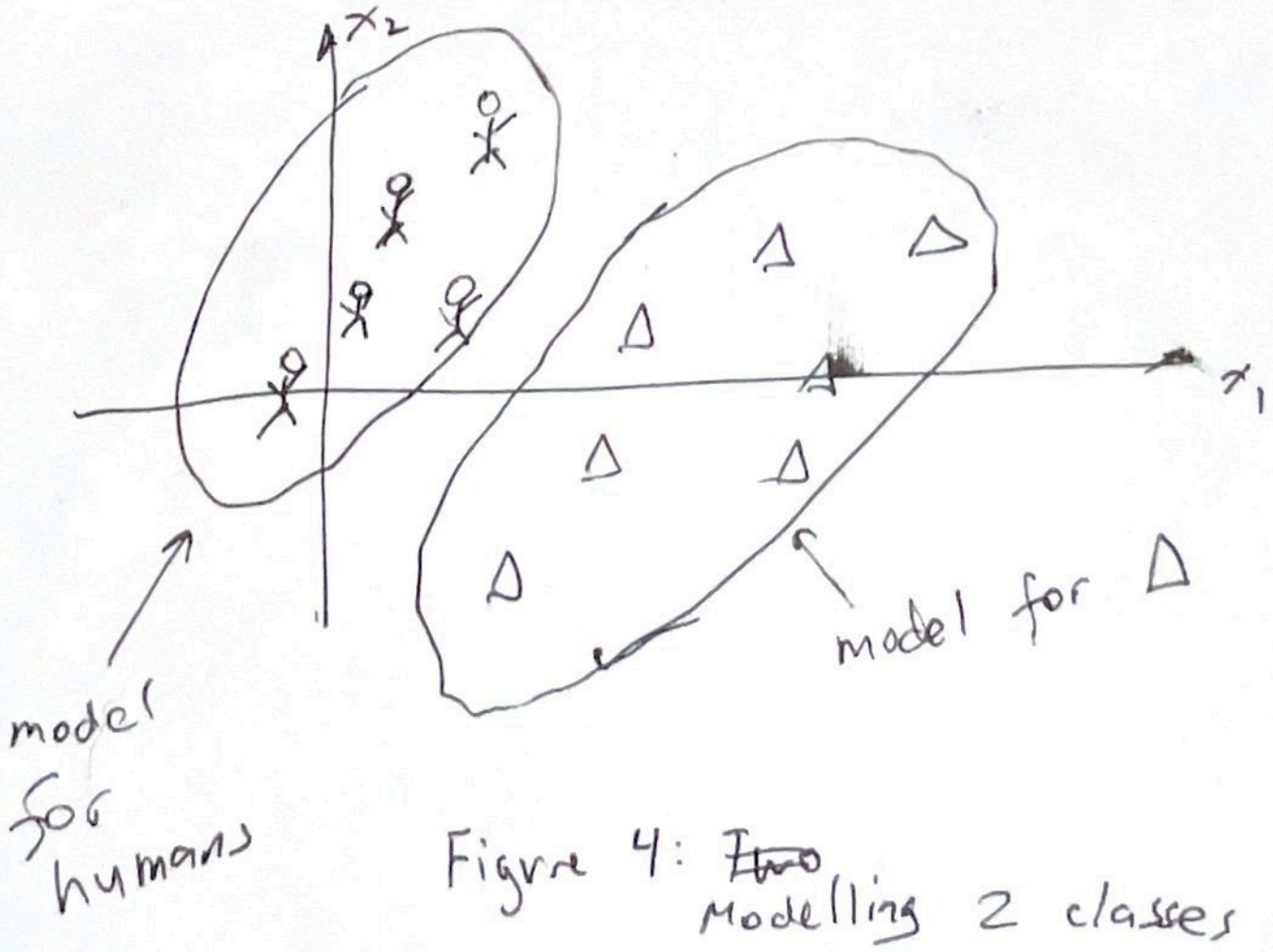


Figure 4: ~~Two~~ Modelling 2 classes

Build a model for each class.

When predicting, compare the sample to each model and pick the one that better matches. ~~a high~~

More accurately use Bayes' Law

$$P(y|\bar{x}) = p(\bar{x}|y)\,P(y)/p(\bar{x}) \quad (5)$$

$p(\bar{x}|y)$: generative model learns this

$P(y)$: ~~estimate prior topically~~ learns prior

$$P(y=1|\bar{x}) = p(\bar{x}|y=1)\,P(y=1)/p(\bar{x})$$

If we are discussing classes by their labels we would write this

$$P(\omega_1|\bar{x}) = p(\bar{x}|\omega_1)\,P(\omega_1)/p(\bar{x})$$

Note for this discussion $\bar{x}$ does not have an add transl 1 as the first element:

$\begin{bmatrix}1\\x_1\\x_2\end{bmatrix}$ ← Not this $\qquad \begin{bmatrix}x_1\\x_2\end{bmatrix}$ ↑ THIS

Gaussian Discriminant Analysis

$\bar{x} \in \mathbb{R}^n$

Assume $p(\bar{x}|y)$ is Gaussian.

Note that $\bar{x}$ is a feature vector. If $\bar{x}|y \sim$ Gaussian then Gaussian must match the dimensionality of $\bar{x}$.

If $\bar{x}$ is 1-dimension, i.e., a scalar, then assume $x|y \sim N(\mu, \sigma^2)$

1-dimensional Gaussian PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (6)$$

(7)

For multivariate Gaussian

$$f_{\bar{X}}(\bar{x}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\bar{x}-\bar{\mu})^T \Sigma^{-1}(\bar{x}-\bar{\mu})\right)$$

$\bar{x}$ is now a random vector

Σ is the covariance matrix $= cov(\bar{X})$

not summation! covariance!

$|\Sigma|$ is the determinant of the covariance matrix

Σ^{-1} = inverse of the covariance matrix

Ex: 2d Gaussian ~~$z = f(x,y)$~~ pdf $f(x_1, x_2)$

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\sigma_{x_1}^2 = 1$ $\sigma_{x_2}^2 = 1$

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$ ← mean of multivariate Gaussian is at the origin of x_1, x_2

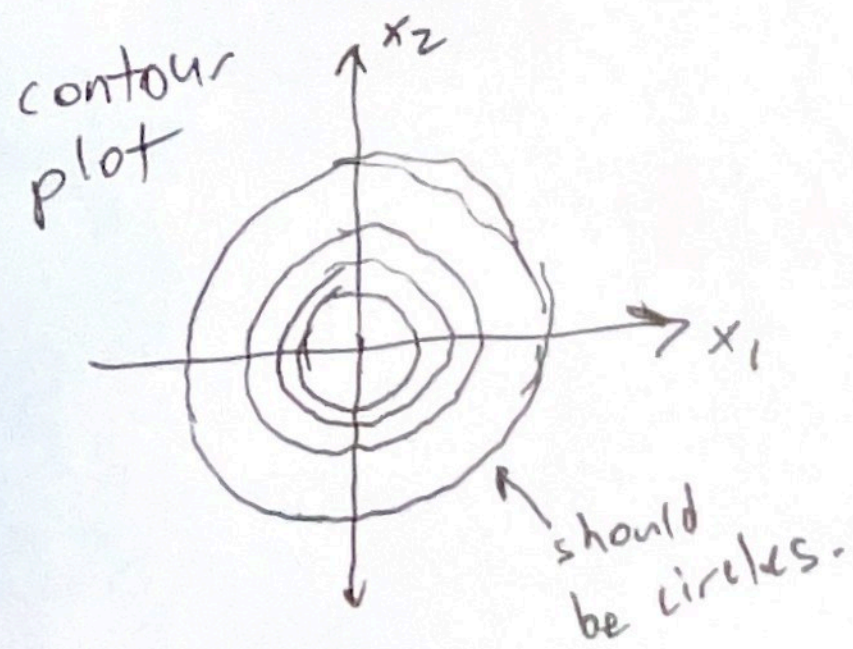


Figure 5: 2d Gaussian PDF with zero covariance

Ex: 2d Gaussian with some covariance

$$\Sigma = \begin{bmatrix} 1 & .8 \\ .8 & 1 \end{bmatrix}$$

Aside Σ is always a symmetric matrix!

Means the covariance of x_1 and x_2 is .8.

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

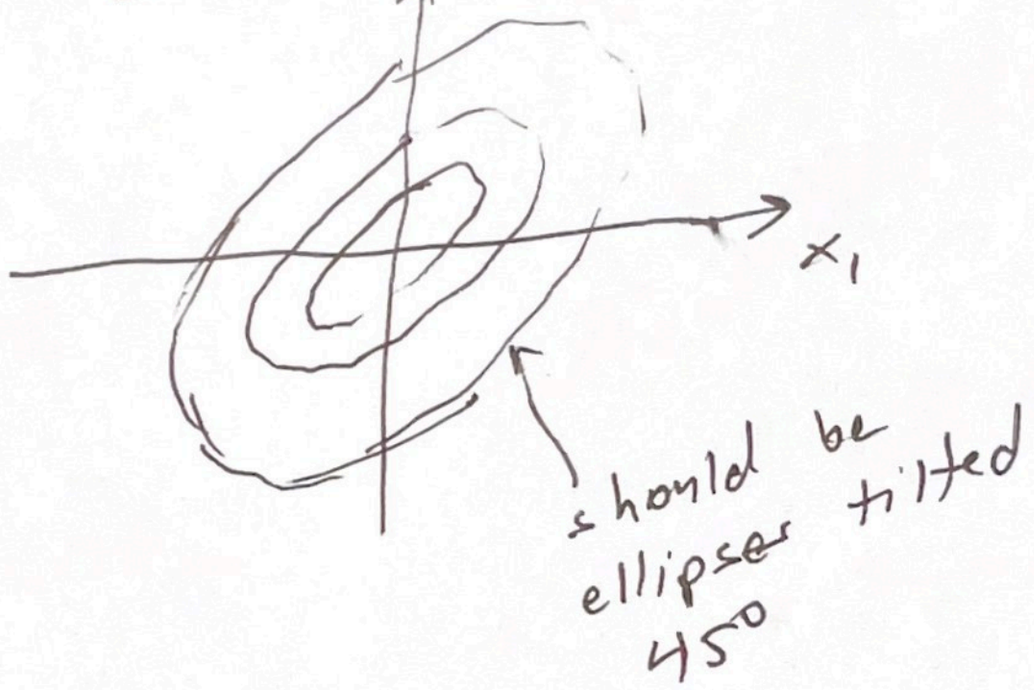


Figure 6: 2d Gaussian PDF with 0.8 covariance

Let $y \in \{0, 1\}$

$$p(\bar{x} \mid y=0) = \frac{1}{(2\pi)^{n/2} |\Sigma_0|^{1/2}} \exp\left[-\frac{1}{2} (\bar{x} - \bar{\mu}_0)^T \Sigma_0^{-1} (\bar{x} - \bar{\mu}_0) \right] \quad (8)$$

$$p(\bar{x} \mid y=1) = \frac{1}{(2\pi)^{n/2} |\Sigma_1|^{1/2}} \exp\left[-\frac{1}{2} (\bar{x} - \bar{\mu}_1)^T \Sigma_1^{-1} (\bar{x} - \bar{\mu}_1) \right]$$

The parameters of the Gaussian are for each class is $(\bar{\mu}_k, \Sigma_k)$. Thus,

k: class k

$$p(\bar{x} \mid y=0) = f_{\bar{\mu}_0, \Sigma_0}(\bar{x}) \quad (9)$$

$$p(\bar{x} \mid y=1) = f_{\bar{\mu}_1, \Sigma_1}(\bar{x})$$

Using Bayes' rule we can then estimate the posterior probabilities of each class

$$P(y=0 \mid \bar{x}) \approx f_{\bar{\mu}_0, \Sigma_0}(\bar{x})\, P(y=0) / p(\bar{x}) \qquad (10)$$

$$P(y=1 \mid \bar{x}) \approx f_{\bar{\mu}_1, \Sigma_1}(\bar{x})\, P(y=1) / p(\bar{x})$$

Minimum error decision rule

assuming Gaussian approximation is accurate

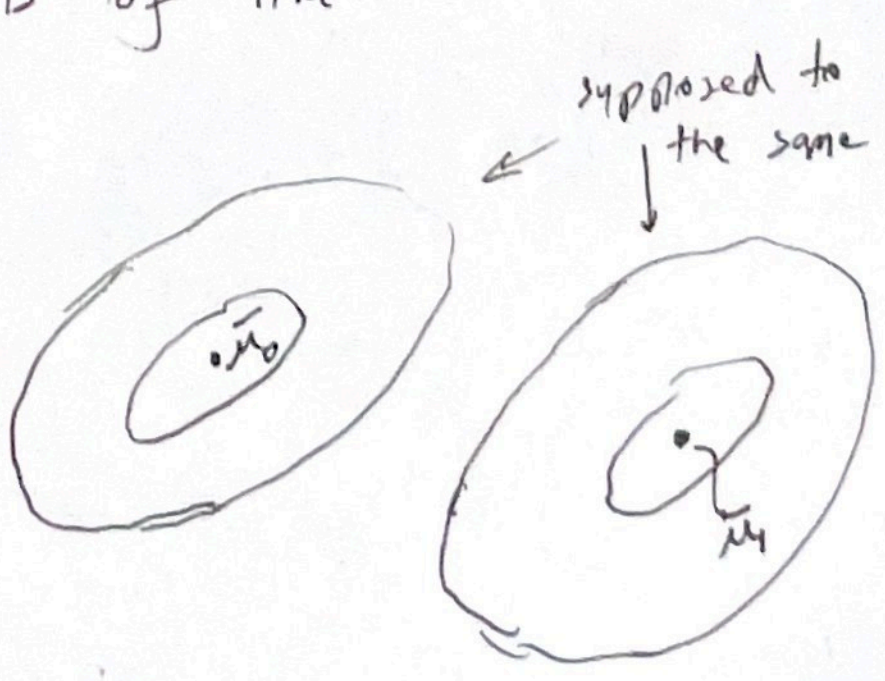
$$y \begin{cases} 0 & \text{if } f_{\bar{\mu}_0, \Sigma_0}(\bar{x})\, P(y=0) > f_{\bar{\mu}_1, \Sigma_1}(\bar{x})\, P(y=1) \\ 1 & \text{otherwise} \end{cases} \qquad (11)$$

(11) arises by cancelling $p(\bar{x})$ from both equations in (10).

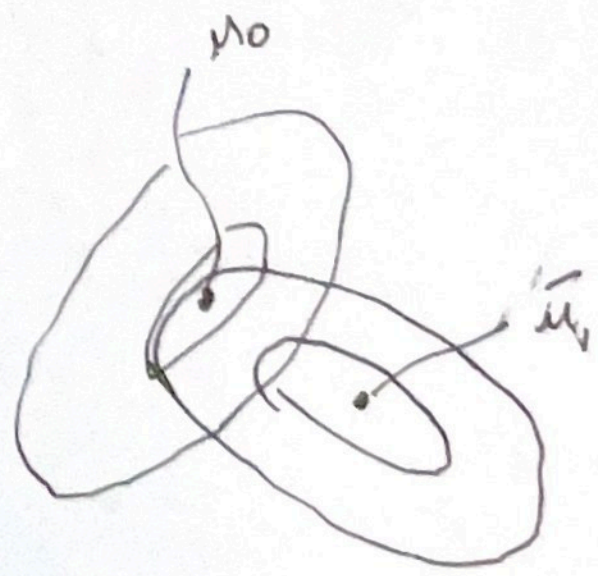
With GDA, we assume $\Sigma_0 = \Sigma_1 = \cdots = \Sigma_{K-1} = \Sigma$

This is known as homoscedasticity.

When we use the same Σ for each class all models of likelihoods share the same covariances



Two 2-d Gaussian PDFs equal Σ

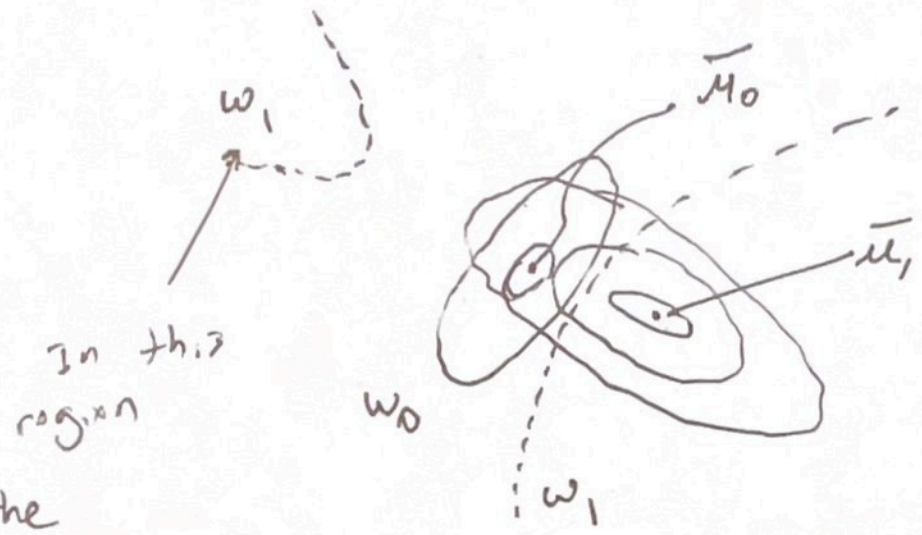


Two 2-d Gaussian PDFs with inequal Σ.

Quadratic Discriminant Analysis (QDA)

QDA allows for separate covariance matrices for each class.

Having separate covariance matrices can result in unexpected decision boundaries, e.g.,



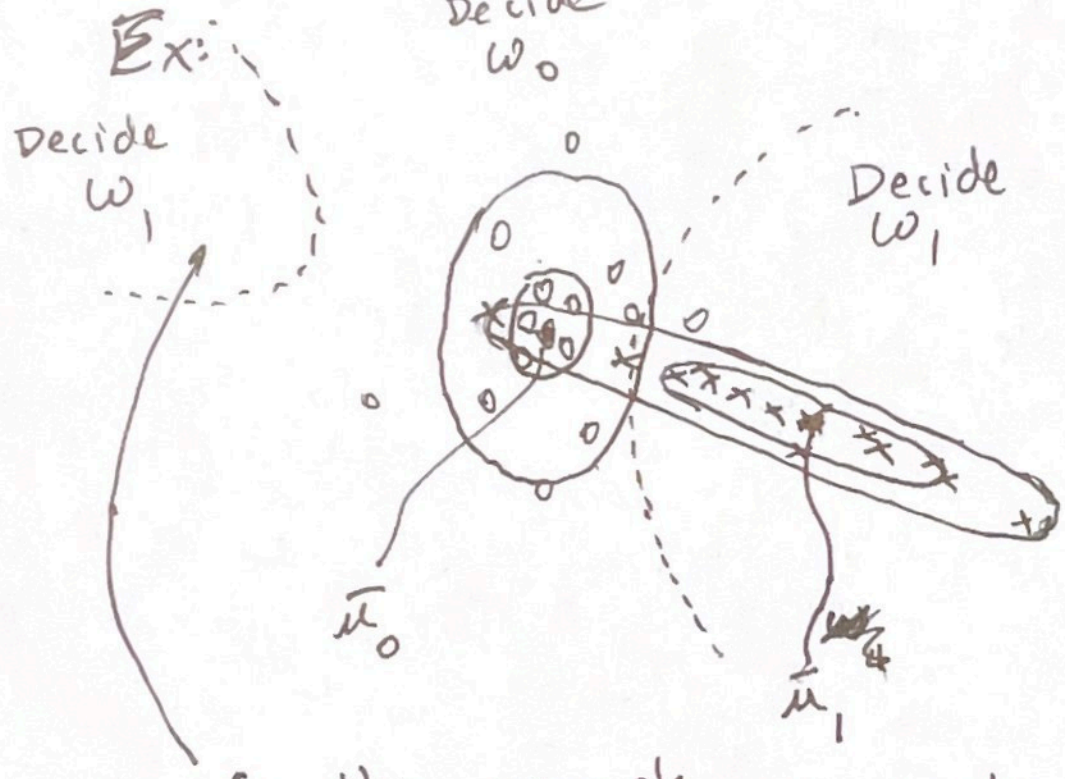
QDA ~~creates~~ can result in decision boundaries that obey any conic (in 2-d):

- ellipse / circle
- hyperbola
- parabola
- line (degenerate conic)

QDA (cont'd)

QDA can result in decision boundaries that are far from the centers of mass of the classes which reside in low probability regions in which zero samples fall.

Ex:



In this example, no samples for class ω_1 (i.e., x's) appear in the region in the upper left hand corner but w/ QDA, modelling ω_1 as a Gaussian results in a decision boundary creating a region for class ω_1 in which no samples in the original data fell.

With GDA, all equal Σ means the decision boundary is always a line. (a hyperplane in n-d)

